

MIC regression

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Preamble

Fit regression models

Complete MIC data (NOS)

```
# apply log to the mics (mix of NOS and 90):
dat_nos_only <- dat_nos_only %>%
  mutate(mic_logged = log2(micnos_1))

# fit a model weighted by inverse sqrt sample sizes to determine residual errors:
# weights in full model
mod_init = lm(mic_logged ~ isolate_yr, dat_nos_only, weights = 1/sqrt(count))
summary(mod_init)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_nos_only, weights = 1/sqrt(count))
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -3.00691 -0.38749 -0.00928  0.00416  2.43792
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -7.5494829  13.5260268  -0.558    0.578
## isolate_yr    0.0007906   0.0067526   0.117    0.907
##
## Residual standard error: 0.6178 on 159 degrees of freedom
## Multiple R-squared:  8.62e-05, Adjusted R-squared:  -0.006203
## F-statistic: 0.01371 on 1 and 159 DF, p-value: 0.9069

# setting weight to median absolute residuals
wts <- data.frame(isolate_yr = dat_nos_only$isolate_yr,
  resid = mod_init$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat_nos_only <- dat_nos_only %>%
  left_join(wts)

## Joining with `by = join_by(isolate_yr)`
```

```
mod_weighted_nos_only <- lm(mic_logged ~ isolate_yr, dat_nos_only, weights = wt)
confint(mod_weighted_nos_only)
```

```
##                2.5 %      97.5 %
## (Intercept) -59.174330679 -3.82369886
## isolate_yr   -0.001150628  0.02654375
```

```
summary(mod_weighted_nos_only)
```

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_nos_only, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.12998 -0.58710  0.00148  0.12213  2.94597
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -31.499015  14.012845  -2.248   0.026 *
## isolate_yr    0.012697   0.007011   1.811   0.072 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8372 on 159 degrees of freedom
## Multiple R-squared:  0.02021,    Adjusted R-squared:  0.01405
## F-statistic: 3.279 on 1 and 159 DF,  p-value: 0.07205
```

```
# this is slope on log-scale:
```

```
coefficients(mod_weighted_nos_only)[2]
```

```
## isolate_yr
## 0.01269656
```

```
2*(coefficients(mod_weighted_nos_only)[2])
```

```
## isolate_yr
## 1.008839
```

```
2*confint(mod_weighted_nos_only)[2,]
```

```
##      2.5 %      97.5 %
## 0.9992028 1.0185690
```

All studies: MIC90

```
# apply log to the mics (mix of NOS and 90):
```

```
dat_agg <- dat_agg %>%
  mutate(mic_logged = log2(mic_90_agg))
```

```
# fit a model weighted by inverse sqrt sample sizes to determine residual errors:
# weights in full model
```

```
mod_init = lm(mic_logged ~ isolate_yr, dat_agg, weights = 1/sqrt(count))
summary(mod_init)
```

```
##
## Call:
```

```
## lm(formula = mic_logged ~ isolate_yr, data = dat_agg, weights = 1/sqrt(count))
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7298 -0.2232  0.0002   0.1207   1.4036
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.141159   7.003181  -0.163   0.871
## isolate_yr  -0.002455   0.003500  -0.702   0.484
##
## Residual standard error: 0.3171 on 211 degrees of freedom
## Multiple R-squared:  0.002327, Adjusted R-squared:  -0.002401
## F-statistic: 0.4922 on 1 and 211 DF, p-value: 0.4837

# setting weight to median absolute residuals
wts <- data.frame(isolate_yr = dat_agg$isolate_yr,
                  resid = mod_init$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat_agg <- dat_agg %>%
  left_join(wts)

## Joining with `by = join_by(isolate_yr)`
mod_weighted_agg <- lm(mic_logged ~ isolate_yr, dat_agg, weights = wt)
confint(mod_weighted_agg)

##              2.5 %      97.5 %
## (Intercept) -28.94219706 20.13498260
## isolate_yr   -0.01305848  0.01149678

summary(mod_weighted_agg)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_agg, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7873 -0.3961 -0.0023   0.0084   3.4116
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.4036072 12.4481080  -0.354   0.724
## isolate_yr  -0.0007808  0.0062283  -0.125   0.900
##
## Residual standard error: 0.7762 on 211 degrees of freedom
## Multiple R-squared:  7.449e-05, Adjusted R-squared:  -0.004664
## F-statistic: 0.01572 on 1 and 211 DF, p-value: 0.9003

# this is slope on log-scale:
coefficients(mod_weighted_agg)[2]

##      isolate_yr
## -0.0007808477
```

```
2**(coefficients(mod_weighted_agg)[2])
```

```
## isolate_yr  
## 0.9994589
```

```
2**confint(mod_weighted_agg)[2,]
```

```
## 2.5 % 97.5 %  
## 0.9909894 1.0080008
```

Now for some plots

Common attributes:

```
YLIM <- range(c(dat_agg$mic_90_agg, dat_nos_only$micnos_1))  
Y_LOWER <- log2(0.0018)  
SIZE_LIMS <- range(c(dat_agg$count, dat_nos_only$count))  
yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)
```

Complete MIC data (NOS only)

```
X_LOWER <- 1965
```

```
# here comes the model again:
```

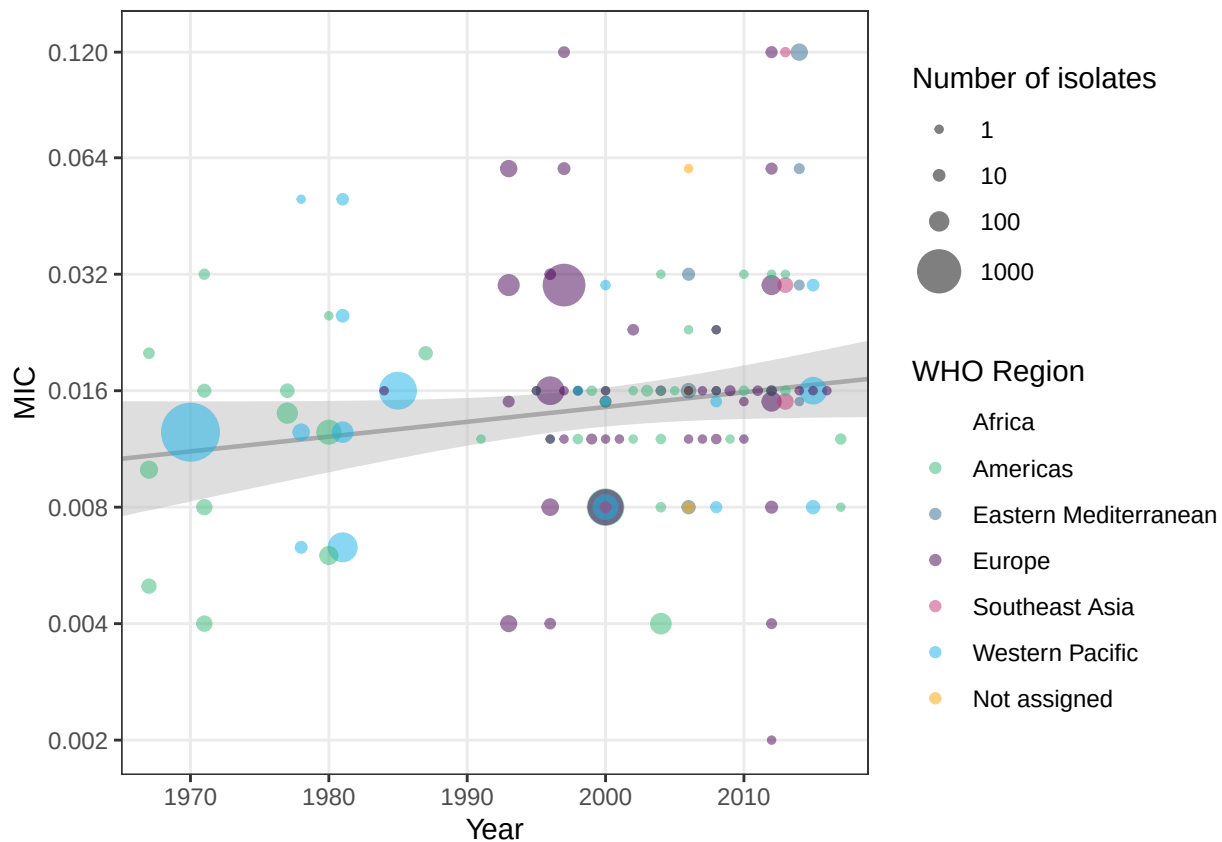
```
newdata <- data.frame(isolate_yr = seq(X_LOWER, max(dat_nos_only(isolate_yr) + 2))  
ci <- predict(mod_weighted_nos_only,  
              newdata = newdata,  
              interval = "confidence") %>%  
  as.data.frame() %>%  
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),  
         isolate_yr = seq(X_LOWER, max(dat_nos_only(isolate_yr) + 2))
```

```
p1 <- ggplot(dat_nos_only %>%  
  filter(!is.na(micnos_1)) %>%  
  group_by(isolate_yr, micnos_1, who_name) %>%  
  summarise(count = sum(count)) %>%  
  mutate(who_name = factor(who_name, levels = c("Africa",  
                                                "Americas",  
                                                "Eastern Mediterranean",  
                                                "Europe",  
                                                "Southeast Asia",  
                                                "Western Pacific",  
                                                "Not assigned")))) %>%  
  arrange(desc(count))) +  
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),  
            alpha = 0.5, fill = "grey", colour = NA) +  
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),  
           colour = "darkgrey", linewidth = 0.8) +  
  geom_point(aes(x = isolate_yr, y = micnos_1,  
                col = who_name,  
                size = count), alpha = 0.5) +  
  scale_color_discrete(name = "WHO Region", type = pal, drop = FALSE) +  
  scale_size_continuous(range = c(1,10), name = "Number of isolates",  
                       breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +  
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
```

```
scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0),
                   breaks = seq(1970, 2010, 10), minor_breaks = NULL) +
labs(x = "Year", ylab = "MIC") +
coord_trans(y = "log2", ylim = YLIM) +
theme_bw()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, micnos_1, and who_name.
## i Output is grouped by isolate_yr and micnos_1.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, micnos_1, who_name))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.
```

p1



All studies: MIC90

```
X_LOWER <- 1927

# here comes the model again:
newdata <- data.frame(isolate_yr = seq(X_LOWER, max(dat_agg$isolate_yr) + 2))
ci <- predict(mod_weighted_agg,
              newdata = newdata,
              interval = "confidence") %>%
as.data.frame() %>%
mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
       isolate_yr = seq(X_LOWER, max(dat_agg$isolate_yr) + 2))
```

```

p2 <- ggplot(dat_agg %>%
  filter(!is.na(mic_90_agg)) %>%
  group_by(isolate_yr, mic_90_agg, who_name) %>%
  summarise(count = sum(count)) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
                                                "Americas",
                                                "Eastern Mediterranean",
                                                "Europe",
                                                "Southeast Asia",
                                                "Western Pacific",
                                                "Not assigned")))) %>%

  arrange(desc(count))) +
geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
  alpha = 0.5, fill = "grey", colour = NA) +
geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
  colour = "darkgrey", linewidth = 0.8) +
geom_point(aes(x = isolate_yr, y = mic_90_agg,
  col = who_name,
  size = count), alpha = 0.5) +
scale_color_discrete(name = "WHO Region",
  type = pal,
  guide = guide_legend(override.aes = list(size = 5),
    ncol = 2)) +
scale_size_continuous(range = c(1,10), name = "Number of isolates",
  breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +
scale_y_continuous(breaks = yticks, minor_breaks = NULL) +
scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0),
  breaks = seq(1930, 2010, 10), minor_breaks = NULL) +
xlab("Year") +
ylab("MIC90") +
coord_trans(y = "log2", ylim = YLIM) +
theme_bw() +
theme(
  legend.box = "horizontal",
  legend.box.just = "top",
  plot.background = element_rect(fill = "transparent", colour = "transparent")
)

```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, mic_90_agg, and who_name.
## i Output is grouped by isolate_yr and mic_90_agg.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, mic_90_agg, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.

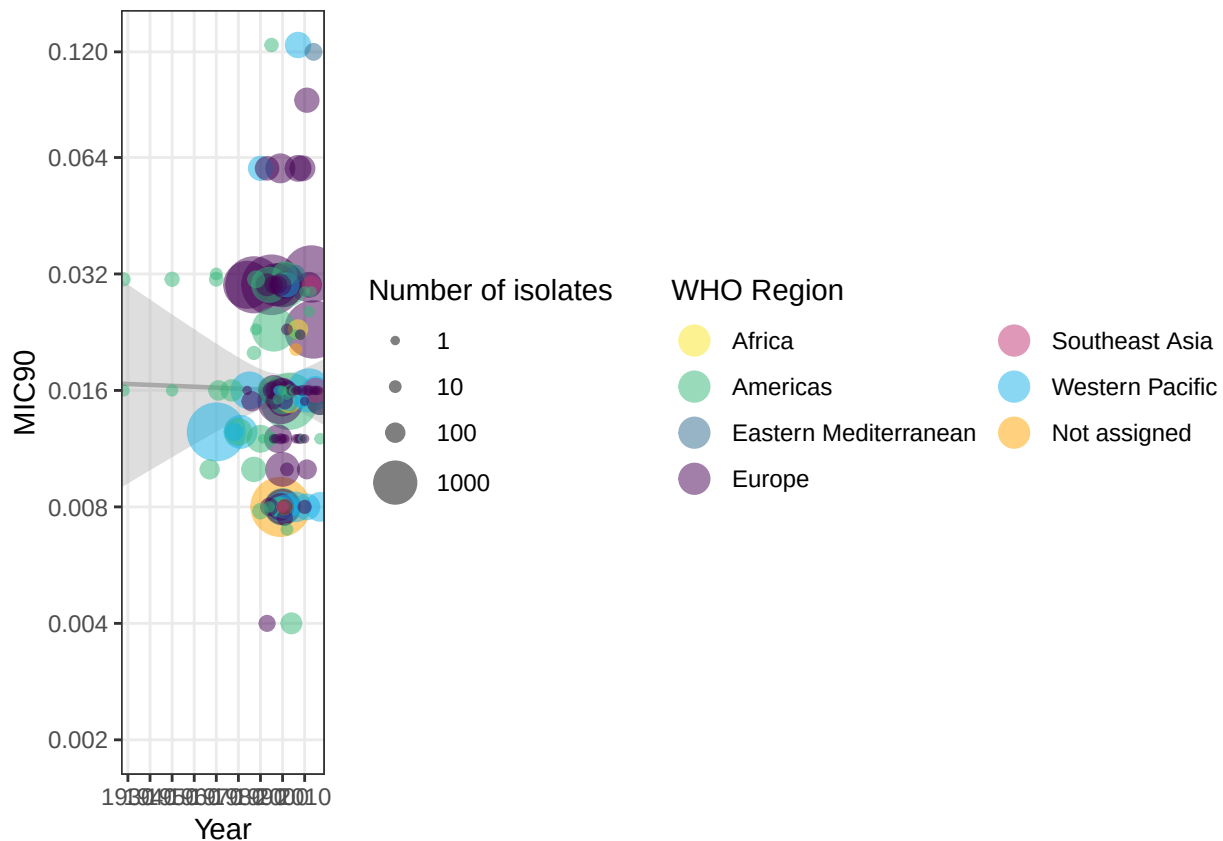
```

p2

```

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

```



Putting it all together:

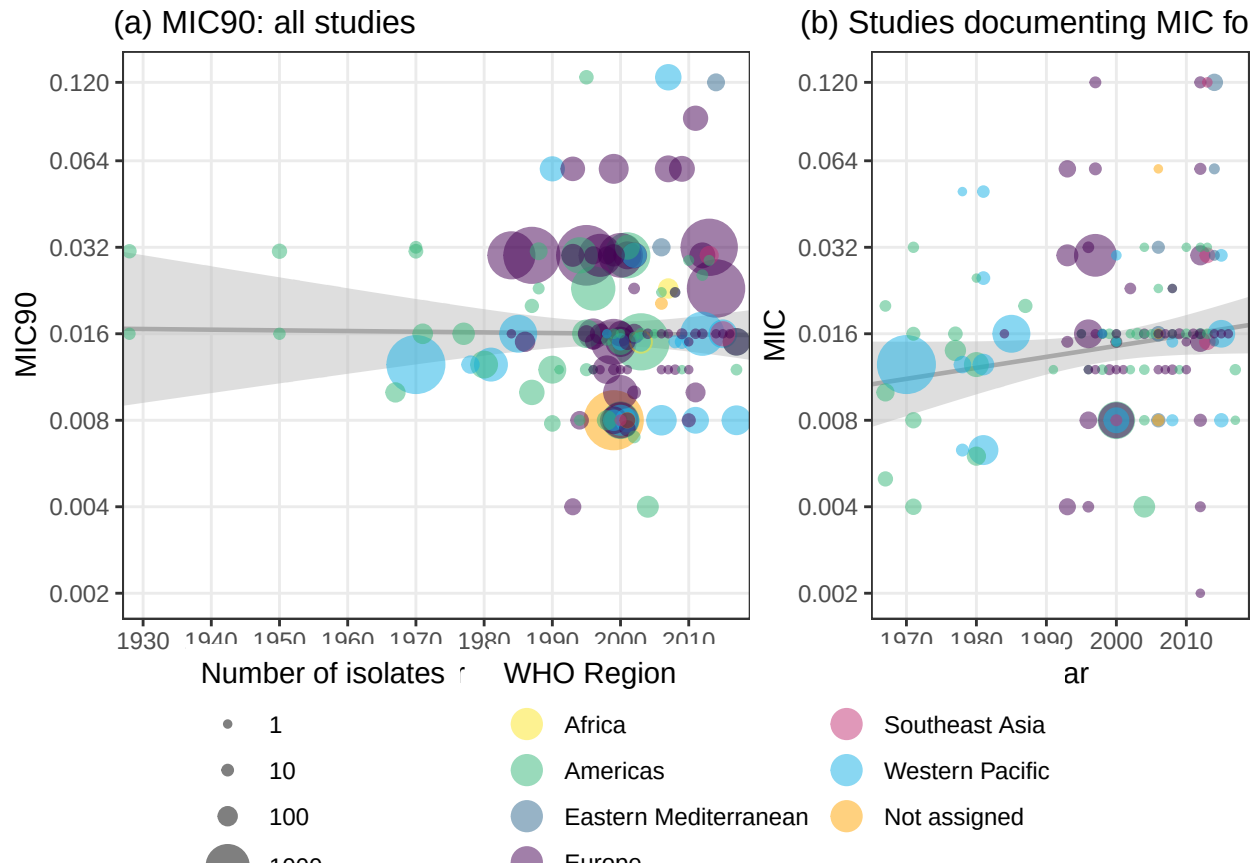
```
p <- plot_grid(p2 + theme(legend.position = "none",
  plot.margin = margin(t = 20, r = 0, b = 0, l = 5, unit = "pt")),
  p1 + theme(legend.position = "none",
  plot.margin = margin(t = 20, r = 0, b = 0, l = 5, unit = "pt")),
  labels = c("(a) MIC90: all studies",
    "(b) Studies documenting MIC for all isolates"),
  label_fontface = "plain",
  rel_widths = c(0.6, 0.4),
  label_x = c(0.15, 0.1), hjust = 0, label_size = 12)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
plot_grid(p, get_legend(p2), ncol = 1, rel_heights = c(0.8, 0.2))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use `return_all = TRUE`.
```



```
ggsave("figs/MIC_regression_twice.png", height = 7, width = 9, bg = "transparent")
```

TODO - report R^2

```
hist_dat <- dat_agg %>% # from Rhys' aggregate file
  filter(!is.na(mic90_1)) %>%
  bind_rows(dat_nos_only) %>% # NOS only
  pivot_longer(cols = c(mic90_1, mic90_2), names_to = "type") %>%
  filter(!is.na(value)) %>%
  mutate(value = round(value, digits = 4)) %>%
  summarise(count = sum(count), .by = c(value, type)) %>%
  mutate(logmic = log2(value)) %>%
  mutate(type = ifelse(type == "mic90_1",
    "Aggregate MIC90 reported only",
    "All MICs reported"))

x_breaks = c(0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

# for labels on bars:
to_lab <- hist_dat %>%
  # only highest ones:
  filter((type == "Aggregate MIC90 reported only" & count > 2000) |
    (type == "All MICs reported" & count > 1000)) %>%
  mutate(nudge_y = ifelse(type == "Aggregate MIC90 reported only",
    1200, 300),
    hjust = ifelse(count %in% c(2727, 2100, 3097), 0, 0.5),
    vjust = ifelse(count %in% c(2727, 3097, 2030), 0.5, 1),
```

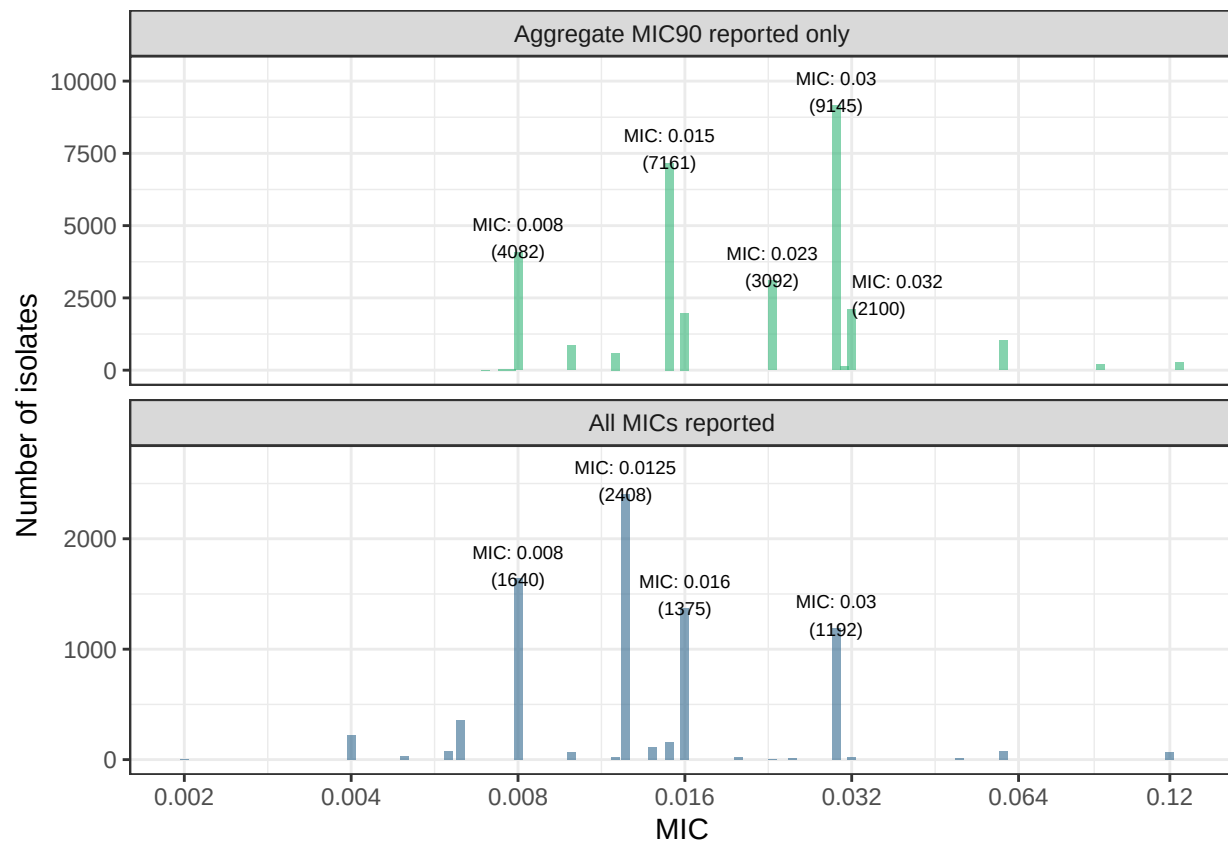
```

    #nudge_x = ifelse(count %in% c(2727, 2100), 0, 0.5),
    lab = ifelse(count %in% c(2727, 3097, 2030),
      paste0("MIC:\n", value, "\n(", count, ")"),
      paste0("MIC: ", value, "\n(", count, ")"))))

ggplot(data = hist_dat) +
  geom_bar(aes(x = logmic, y = count,
    fill = type),
    alpha = 0.6,
    stat = "identity",
    width = 0.05) +
  # here are labels on top of higher bars:
  geom_text(aes(x = logmic, y = count, label = lab),
    data = to_lab,
    hjust = to_lab$hjust,
    vjust = to_lab$vjust,
    nudge_y = to_lab$nudge_y,
    #nudge_x = to_lab$nudge_x,
    size = 2.5) +
  facet_wrap(~ type, ncol = 1, scales = "free_y") +
  scale_x_continuous(breaks = log2(x_breaks),
    labels = x_breaks) +
  scale_fill_manual(values = pal[2:3]) +
  ylab("Number of isolates") +
  xlab("MIC") +
  theme_bw() +
  theme(legend.position = "none")

```

Warning: `position\_stack()` requires non-overlapping x intervals.



```
ggsave("figs/all_mics.png", height = 6, width = 6)
```

```
## Warning: `position_stack()` requires non-overlapping x intervals.
```